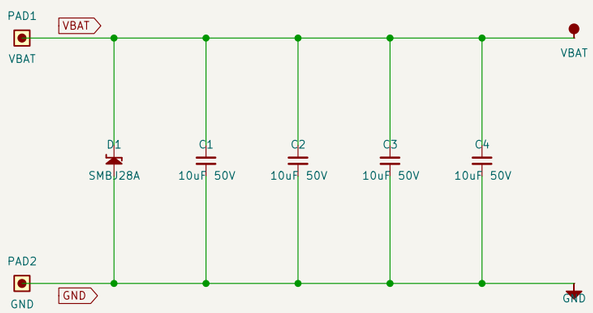
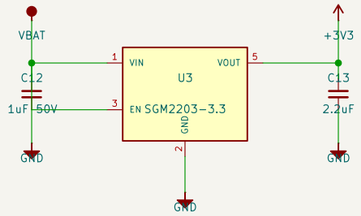
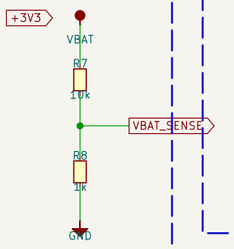
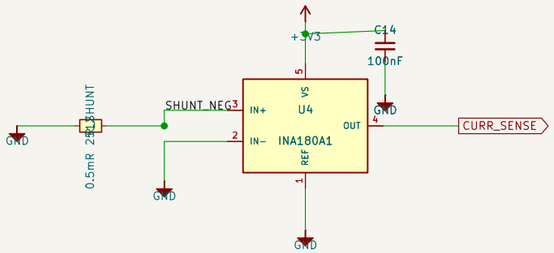
SECTION 1: POWER ENTRY, REGULATION & CURRENT SENSING

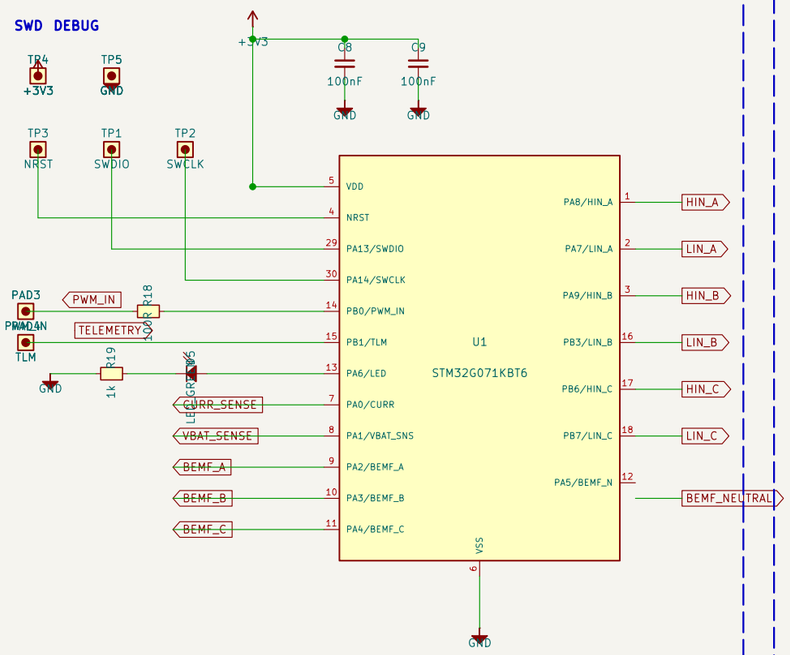




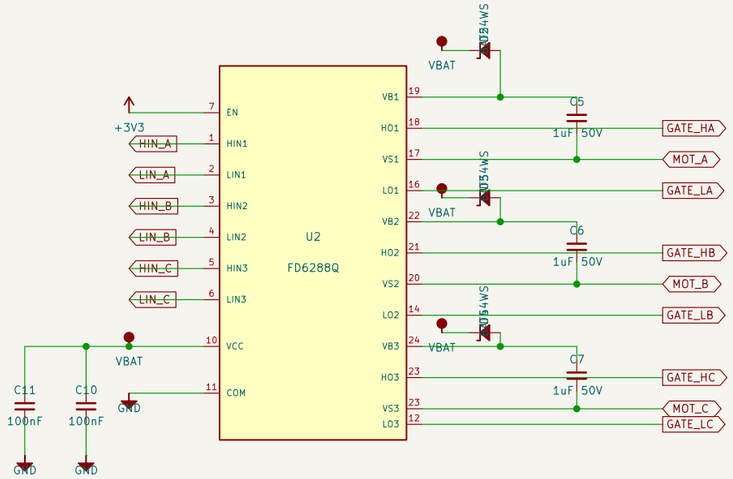




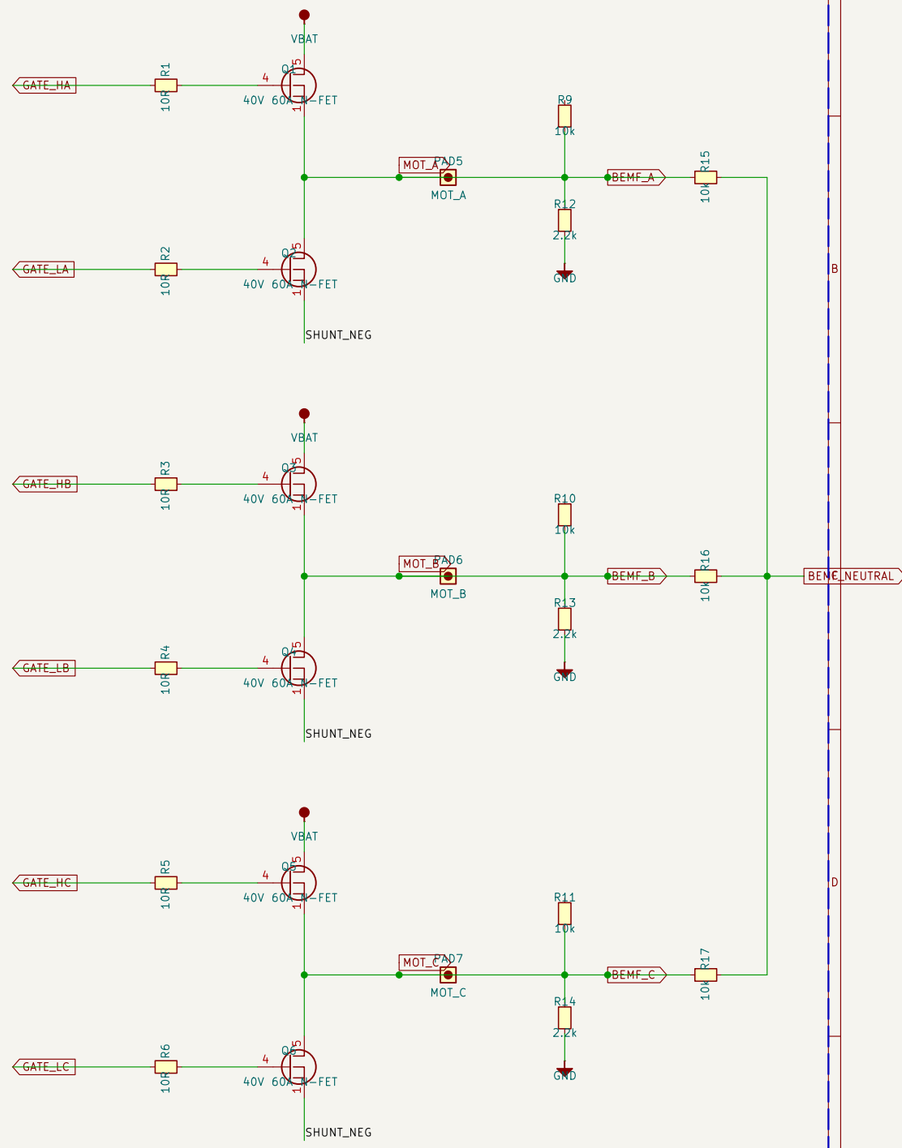
SECTION 2: MICROCONTROLLER & PERIPHERALS (STM32G071)



SECTION 3: 3-PHASE GATE DRIVER (FD6288Q)



SECTION 4: 3-PHASE INVERTER BRIDGE & BEMF SENSING



Hardware Architecture: INA180 Current Amp, SGM2203 3.3V LDO, BEMF Divider Network
Inverter: 6x 40V 60A N-FETs | Rating: 3S-6S LiPo (9-25.2V) 35A Cont / 45A Burst
MCU: STM32G071KBT6 (QFN-32) | Driver: FD6288Q (QFN-24)

Autonomous Drone Systems

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Title: CUSTOM 3S-6S 35A/45A SINGLE BLDC ESC

Size: A3	Date: 2026-11-11	**Rev: v1.1**
KiCad E.D.A. 10.0.0		Id: 1/1